

The Limits of Blockchain Forensics: Deconstructing the Mythologies of Automated Governance

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Abstract

Regulators currently mandate automated scrutiny for a \$24.2 billion cryptocurrency illicit flow problem, despite a \$2 trillion global laundering industry. This is underpinned by a multi-billion dollar blockchain intelligence industry that markets high-complexity AI for automated governance and market prediction. This paper evaluates the empirical and operational limits of public illicit labels in blockchain forensics across four core questions: the marginal utility of structural versus behavioral data (RQ1), the predictive power of illicit flows for market movements (RQ2), the operational feasibility of standalone automated alarms under extreme rarity (RQ3), and the microstructure detection of DEX wash trading (RQ4). Employing a multi-stage methodology, we benchmark Graph Neural Networks (GNNs) against simple tree ensembles using degree-matched protocols across three wallet benchmarks, subject risk-weighted illicit-flow signals to a battery of thirty statistical tests across major cryptocurrencies, introduce a rolling calibration stability test to expose fleeting signals, quantify operational review workloads under a “Base-Rate Deployability” framework, and assess a behavioral DEX screening pipeline. Our findings systematically deconstruct the industry’s “Complexity Trap.” First, tree ensembles are competitive with evaluated GNNs, while the utility of structural features is network-specific; on the flatter Bitcoin graph, structural-only Random Forest performance ($F1 \approx 0.893$) exceeds behavioral-only performance (0.844), but in top-heavy networks behavioral features dominate. Second, every apparent market signal for price, volatility, or liquidity collapses to a broad null once required to maintain directional stability across rolling calibrations; observed in-sample associations are sensitive to price feeds, look-ahead leakage, or evaluation periods. Third, we quantify the operational impact of the base-rate fallacy. At an estimated attacker-contract prevalence of 2.8–8.1 per million deployments (central estimate $\approx 4.4 \times 10^{-6}$), achieving 90% recall imposes a projected operational review burden of approximately 40,700–116,700 benign reviews per catch conditional on prevalence and decision thresholds, demonstrating that standalone automated alarms exceed practical review capacity. Fourth, our behavioral DEX screening pipeline isolates pool-dominating, net-flat trader activity consistent with wash trading, formalized via a candidate-associated “Wash-Discount” metric, though requiring contextual validation. Finally, we document an approximate two-year activity half-life for public labels in the examined Ethereum cohort, reflecting operational address lifecycle decay rather than label invalidity. We conclude that current mandates for automated blocking ignore the mathematical reality of the base-rate fallacy. Stakeholders must pivot from viewing illicit labels as automated “kill-switches” toward utilizing them as triage tools for prioritizing human review, acknowledging that operational deployability is strictly governed by base-rate prevalence, operating thresholds, and institutional review capacity.

Introduction

The United Nations Office on Drugs and Crime (UNODC) estimates that the amount of money laundered globally in one year is 2% to 5% of global GDP, roughly \$800 billion to \$2 trillion, stemming primarily from drug trafficking, tax evasion, and human trafficking. Of that, crypto-specific illicit flows account for less than 1% of total transaction volume—estimated at approximately \$24.2 billion in 2023 (Chainalysis 2024; TRM Labs 2024). Yet regulators, attracted to “security theater” that leans on the latest technological fads, demand a level of automated scrutiny for this \$24.2 billion problem that they have not achieved for the larger \$2 trillion global laundering problem.

Estimates of illicit cryptocurrency activity are small relative to estimates of global money laundering, but public ledgers make cryptocurrency activity unusually observable. This observability has encouraged the extensive use of public address labels and automated scoring systems in compliance and forensic analysis. The central question is whether this automated focus on illicit cryptocurrency flows, and the associated enforcement cost, is effective and evidence-based. This paper demonstrates that primary automated methodologies marketed for banks, governments, and exchanges face significant mathematical and conceptual constraints that limit their utility as autonomous controls.

The commercial blockchain intelligence industry (e.g., Chainalysis, TRM Labs, Elliptic) often emphasizes advanced analytics and AI-driven capabilities to address compliance challenges. However, a closer examination of high-profile interdictions reveals that enforcement successes are frequently driven by simple behavioral heuristics, operational errors by perpetrators, and conventional investigative work (such as subpoenaed server logs and exchange records) rather than autonomous on-chain algorithms alone. For example, the 2016 Bitfinex hack, resulting in the largest seizure in history (roughly \$3.6 billion), is widely cited in commercial narratives; yet the suspects were caught not by an autonomous algorithm, but because they stored unencrypted private keys in a plain-text file on a centralized cloud provider—a classic case of traditional police work following a human mistake. Similarly, the capture of James Zhong, who hid 50,000 stolen Silk Road Bitcoins for a decade, was triggered by a manual error in linking a “dirty” wallet to a KYC-compliant exchange.

These cases highlight an important distinction between marketing narratives and operational reality. Blockchain-forensics research leans heavily on public illicit labels, such as Etherscan tags, Forta alerts, and the *Elliptic Dataset* (Elliptic 2019). Researchers and compliance vendors build detectors on these labels to detect illicit wallets, construct market signals, and prioritize alerts. A detector

trained on these labels will routinely report high aggregate classification scores, but the prior question is almost always skipped: Are these labels and derived scores actually informative and reliable under task-appropriate evaluation?

Public case accounts illustrate that blockchain tracing is typically one component of a broader investigative process involving exchange records, conventional evidence, and analyst judgment. Because commercial platforms incorporate proprietary off-chain intelligence, private entity clusters, and specialized investigator workflows, their end-to-end commercial systems are not directly benchmarked here. Rather, our empirical tests evaluate the core algorithmic claims regarding the incremental information in public labels, open graph learning architectures, and automated screening models.

Our research addresses four core questions:

- **RQ1: Network-Specific Structural Value vs. Behavior.** When a detector identifies a wallet as illicit, is it reacting to the complex network structure (the graph) or simply to how the wallet behaves (transaction frequency, recency, age), and how does network concentration moderate this relationship?
- **RQ2: Market Prediction.** Does a signal derived from illicit flows actually predict next-day price, volatility, or liquidity, or is the perceived “alpha” merely a data artifact?
- **RQ3: Standalone Alarms.** Can such a detector be deployed as a standalone alarm given the mathematical reality of how rare attackers actually are in the general population, conditional on prevalence, thresholds, and review capacity?
- **RQ4: Microstructure Manipulation.** Can wash trading on decentralized exchanges (DEXs) be screened from trading behavior when standard pool-level tests do not carry over?

Literature Review: Theoretical Anchoring & Three Mythologies

Theoretical Foundations

To establish a rigorous conceptual foundation, this study bridges Information Systems (IS), Financial Economics, and Algorithmic Forensics across four core theoretical pillars:

1. **Information Asymmetry (Akerlof 1970):** In on-chain environments, illicit actors deliberately obfuscate transaction trails through peeling chains, mixers, and multi-hop hops.

Public labels attempt to bridge this lemon-market asymmetry, but their opacity, staleness, and latency often create false signals.

2. **Design Science Research (Hevner et al. 2004; March and Smith 1995):** We evaluate blockchain forensics tools not merely as passive predictive algorithms, but as *designed artifacts* whose utility must be judged by their operational performance and cost-effectiveness in real-world deployment environments.
3. **Informational Efficiency & Market Hypotheses (Fama 1970; Grossman and Stiglitz 1980):** Public blockchain transactions are broadcast on open ledgers. If on-chain flows contained genuine predictive signals for future returns, competitive market participants would arbitrage away the predictability, leaving zero residual out-of-sample alpha.
4. **Market Microstructure Theory (Kyle 1985):** Informed trading impacts liquidity, bid–ask spreads, and order flow imbalance (OFI). We examine whether illicit money movements create structural order imbalance at granular timescales.

The Technical Mythology: Graph Learning vs. Behavioral Features

The “Technical Mythology” asserts that the inherent graph structure of distributed ledgers necessitates high-complexity, relational models to identify illicit activity. This premise was codified by the introduction of the *Elliptic Dataset* (Elliptic 2019), the primary benchmark for node classification in financial forensics. Influential work by Weber et al. (2019) established Graph Convolutional Networks (GCNs) as the standard for capturing relational nuances in Bitcoin flows. This was further extended by Pareja et al. (2020), who argued for “evolving” graph structures (EvolveGCN) to account for temporal dynamism.

Industry practitioners have operationalized this “graph-first” approach through proprietary tools, marketing “blackbox” AI as an indispensable necessity for “Know Your Transaction” (KYT) compliance (Chainalysis 2024). However, this reliance on structural complexity ignores the efficacy of simpler behavioral heuristics. Recent machine-learning benchmarks suggest that simple tree-based ensembles often match or outperform GNNs on tabular graph data. We test the hypothesis (**H1**) that structural features drive detection by comparing feature ablations and model families under matched evaluation procedures. The results show competitive tree/GNN performance and demonstrate that structural-feature utility is coin-specific, depending heavily on network concentration.

The Informational Mythology: On-chain Flows and Market Prediction

The “Informational Mythology” posits that the transparency of the blockchain enables superior market prediction, assuming “Whale” movements or net flows provide leading signals for market volatility and price trajectory. Large-scale data providers like *Glassnode* (Glassnode 2024) and *CryptoQuant* (CryptoQuant 2024) have operationalized this, treating metrics like *Exchange Net Flow* as “received wisdom” indicators of imminent sell pressure.

Academic literature frequently references Griffin and Shams (2020), who argued that Tether (USDT) flows were timed to manipulate Bitcoin prices. Similarly, Liu and Tsyvinski (2021) established that cryptocurrency returns are driven by network-specific factors like user adoption. Yet, as noted by Makarov and Schoar (2020), while arbitrage opportunities and flows exist, they are often constrained by capital controls rather than lack of information, suggesting that many perceived “predictive” signals are noise already priced in by efficient market actors. We test the hypothesis (**H2**) that illicit flows lead the market, utilizing a battery of thirty statistical tests alongside a rolling calibration stability test to expose signals that appear statistically significant in full-sample regressions but fail across temporal windows.

The Operational Mythology: Rare Events and the Base-Rate Fallacy

The “Operational Mythology” claims that standalone, automated blocking of transactions is a viable compliance standard. This is reinforced by global regulatory bodies; for instance, the *Financial Action Task Force (FATF)* (Financial Action Task Force 2021) mandates that Virtual Asset Service Providers (VASPs) utilize automated tools to monitor transactions and comply with “Travel Rule” standards (Financial Crimes Enforcement Network 2020b, 2020a).

This regulatory mandate, however, ignores the mathematical reality of the *Base-Rate Fallacy*. As established in the foundational work by Axelsson (2000), the operational efficacy of any intrusion-detection or alarm system is limited not merely by its classification accuracy, but by the rarity of the event it seeks to detect. In high-noise environments where illicit actors comprise a negligible fraction of total transactions (approximately 2.8–8.1 per million deployments), even a “state-of-the-art” system with strong ROC-AUC discrimination generates an operationally burdensome volume of false positives. This creates a severe “deployment gap” where, as Brumme et al. (2021) note, the friction of reviewing false alarms in AML systems prevents autonomous deployment, leading to a state of

“security theater” where tools are deployed for regulatory optics rather than genuine interdiction. We test (**H3**) by combining operating-point error rates with an estimated attacker-contract prevalence to report benign reviews per true positive as a conditional workload projection governed by prevalence, decision thresholds, and review capacity.

Methodology

This section describes four related research questions with task-specific data, methods, and outputs. Figure 1 shows the end-to-end six-stage research architecture, from wallet detection and causal signal construction to market-forecasting tests, additional signal analyses, operational contract screening, and DEX wash-trade screening.

Wallet detection and negative sampling (RQ1)

The wallet detector reads sixteen features per wallet: seven structural and nine behavioral. Structural features describe where a wallet sits in the transaction graph (log degree, log PageRank, average neighbor degree, k-core, betweenness, clustering, closeness); behavioral features describe what a wallet does (log incoming and outgoing value and count, an activity ratio, and four temporal features for duration, recency, age, and burst).

We compare a Random Forest, a gradient-boosted tree (XGBoost), GraphSAGE, a plain Graph Attention Network (GAT), and an attention-based Triple-Attention model, as well as a linear Logistic Regression baseline. Two choices ensure fair comparison: 1. **Degree-Matched Negative Sampling:** On Shiba Inu and Bitcoin, every illicit wallet is paired with a normal wallet of similar transaction degree, preventing an artificial classification victory achieved simply by flagging high-volume wallets. DOGE uses curated negatives, providing a useful contrast. 2. **K-Fold Protocol:** Evaluation uses five-fold cross-validation on the degree-matched balanced set. The detector architecture is modular, ensuring it can be evaluated independently from downstream market-signal construction.

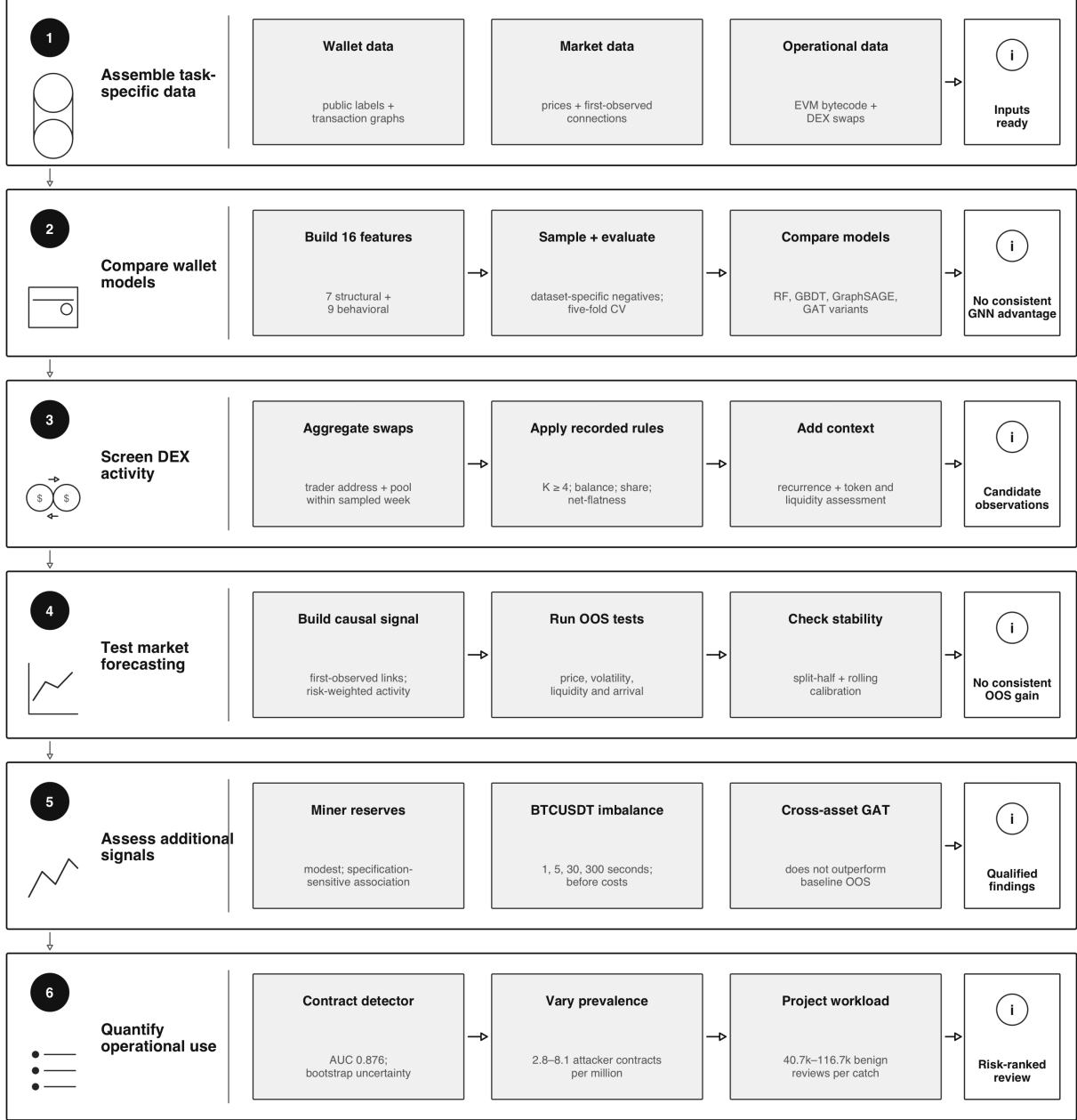


Figure 1: Six-stage research architecture. Task-specific data support the wallet-model comparison, DEX candidate screening, market-forecasting tests, additional signal analyses, and contract-screening workload assessment. Each row reports its own inputs, procedures, and qualified output.

Constructing the daily illicit-flow signal (RQ2)

The detector assigns each wallet a risk score between zero and one. We treat these scores as ranking measures rather than assuming they are calibrated probabilities. From these scores, we construct a daily metric capturing suspicious network activity: a risk-weighted measure of newly observed wallet connections, accompanied by the share of new connections associated with illicit labels. The metric measures connection activity rather than transferred monetary volume.

On each day, we include connections strictly on their first-observed date and weight them by available wallet risk scores. Lifetime activity is not distributed retrospectively across earlier dates. This timing safeguard prevents future connection activity from leaking into an earlier day’s aggregate. A fully point-in-time interpretation would additionally require timestamped model versions, labels, and feature inputs; those elements are not established by this safeguard alone.

The Look-Ahead Leak Avoided

The tempting mistake in on-chain signal generation is to distribute a connection’s lifetime activity backward across every day it was active, from first observation to last observation. Because spreading activity that way requires knowing the connection’s last-seen date—which sits in the future relative to the transaction date—it quietly injects look-ahead information into historical series.

We tested signal construction both ways to evaluate vulnerability to this leak. Under the leaky version, Dogecoin produced an apparently favorable out-of-sample volatility forecasting gain (MSE ratio ≈ 0.934). When the series was rebuilt causally using only first-observed connections, the result collapsed and vanished (MSE ratio ≈ 1.044), proving that the apparent discovery was entirely an artifact of look-ahead leakage.

Testing the illicit-flow to price link (RQ2)

Our empirical question is whether today’s illicit-flow signal helps forecast the next day’s price or volatility, controlling for total on-chain activity, trading volume, recent volatility, Bitcoin’s own volatility as a market-wide factor, and macroeconomic controls.

To ensure that a null result is robust and well-powered rather than an artifact of test selection, we subject the signal to roughly thirty statistical tests:

- **Granger Causality:** Lags 1 through 10 days, in both directions, alongside weekly aggregations.
- **Out-of-Sample HAR Realized-Volatility Forecasting:** Expanding window evaluations judged by the Clark-West test and a circular-block permutation null.
- **Nonlinear & Tail Dependencies:** Tail, jump, downside-semivariance tests, and Transfer Entropy.
- **Causality and Forecast Comparison:** Toda–Yamamoto causality tests for integrated series and Hansen’s Superior Predictive Ability (SPA) forecast-comparison test.
- **Frequency Domain & Deep Learning:** Breitung-Candelon frequency-domain tests, multi-coin panels, and a Temporal Fusion Transformer (TFT).

Forecasting exercises use expanding out-of-sample windows. Benjamini-Hochberg false discovery rate (FDR) multiplicity corrections and minimum-detectable-effect power analyses are applied to their documented test families.

Going beyond price to liquidity, spreads, and exchange-arrival

We extended the evaluation target set to market microstructure:

1. **Liquidity:** Amihud illiquidity ratio ($\frac{|r_t|}{\text{Volume}_t}$).
2. **Trading Spreads:** Corwin-Schultz high-low bid-ask spread estimator.
3. **Exchange Arrival Pressure:** Illicit inflow directly to exchange deposit wallets (proxies for imminent selling pressure) at daily and hourly frequencies.

The Sub-Period Stability Gate

To evaluate sub-period stability, we subject apparent market signals to a rolling calibration framework, comparing the rolling-window parameter distribution against two benchmark calibrations: a shuffled-null baseline and an injected-signal series. A genuine economic signal should maintain directional consistency across temporal windows, approaching the injected benchmark (≈ 0.93 positive estimates). For the Ethereum liquidity relationship, the observed series yields a positive rolling-window fraction of approximately 0.68, which clusters closer to the uninformative shuffled null (0.49) than to a persistent directional relationship. These rolling calibration diagnostics assess

whether an empirical relationship exhibits stable predictive power or whether in-sample statistical significance reflects transient window variation.

Attacker-contract screening (RQ3)

When a new smart contract is deployed, the detector reads its raw EVM bytecode to predict whether the contract is an exploit before execution. We extract push-aware opcode n-grams, stepping over pushed data literals. The classifier is an XGBoost model evaluated under a strict temporal split: trained on pre-2023 contracts and tested on 2023–2024 contracts. Exact hash duplicates and near-duplicate bytecode clusters are strictly segregated to prevent template leakage across splits. The full-anchor evaluation yields AUC 0.8762; a separate near-duplicate cluster-aware robustness exercise yields AUC 0.8885. This design evaluates retrospective screening and does not demonstrate live interception prior to contract execution.

Base-Rate Deployability Accounting

We complement standard ROC-AUC and F1 scores with an operational cost calculation. An illustrative decision scenario assigns \$3,000,000 to a missed exploit and \$100 to an analyst review. The decisive operational metric is *benign reviews per catch* (false positives per true positive):

$$\text{reviews-per-catch} = \frac{1-p}{p} \cdot \frac{\text{FPR}_{\text{at recall}}}{\text{recall}}, \quad (1)$$

where p is the empirical base rate (prevalence) of attacker contracts among all deployed contracts. Total reviewed alerts per catch adds the true-positive alert to this quantity. This metric provides a conditional projection of review workload given target prevalence and chosen operating thresholds.

We estimate attacker-contract prevalence empirically by comparing reported exploit contracts against total deployment counts:

$$p \approx \frac{150 \text{ known exploit contracts}}{37.5\text{M total deployments}} \approx 4 \times 10^{-6},$$

with a central estimate of $p = 4.4 \times 10^{-6}$ and a sensitivity band of 2.8×10^{-6} to 8.1×10^{-6} (roughly 3 to 8 attackers per million deployments). This prevalence is specific to attacker contracts and should not be transferred to general wallets or transaction volume.

Entity-Level Gating for DEX Wash Trading (RQ4)

To address wash trading on decentralized exchanges where standard pool-level volume tests fail to carry over, behavioral analysis must evaluate whether trading activity exhibits non-economic recycling of inventory. While a comprehensive conceptual framework identifies multiple behavioral dimensions (such as closed-loop routing, inventory neutrality, fee incurrence, offsetting order flow, and pool volume concentration), operational screening on public ledgers relies on address-level screening criteria that do not presuppose complete multi-address entity resolution.

Operationally, we aggregate swaps by trader address and pool within each sampled week. Candidate screening implements four consistent screening rules: 1. **Activity threshold:** Minimum buy/sell activity count $K \geq 4$. 2. **Directional balance:** Buy/sell count balance $\frac{\min(\text{buys}, \text{sell})}{\max(\text{buys}, \text{sell})} \geq 0.60$. 3. **Pool volume share:** Address swap-count share of pool activity ≥ 0.30 . 4. **Inventory neutrality:** Net-to-gross flow fractions below 0.10 in both token legs, restricting pools to no more than 2,000 swaps.

Sustained-candidate assessment examines recurrence across at least two of three within-window sub-periods, followed by external token-legitimacy and pool liquidity assessment. Here K is an aggregate activity threshold. This method identifies candidate trader-address observations consistent with volume manipulation; it flags candidate activity for contextual verification rather than proving multi-address collusion or matched round-trip sequences.

Dataset and Experimental Setup

- **Cryptocurrencies:** Wallet comparisons cover Shiba Inu (SHIB), Dogecoin (DOGE), and Bitcoin (BTC). The primary illicit-flow forecasting table covers SHIB, Ethereum (ETH), and DOGE over 2021–2024, extended through 2026. Boundary generalization tests are conducted on World Liberty Financial (WLFI) and the USD1 stablecoin. Supporting analyses use their documented asset-specific periods.
- **Price Feeds:** yfinance, Binance.US, and CoinGecko (independent triple-feed validation).
- **Forensic Datasets:** Elliptic++ Bitcoin dataset, Etherscan/Forta blacklists, curated Dogecoin scam addresses, DefiHackLabs, Forta, LookAhead, and SEASONED exploit repositories. Cross-domain benchmarks include NSL-KDD (network intrusion), Spambase (email spam), and ClaMP (PE malware).

- **Reproducibility:** Seed-frozen pipelines (seed=42) evaluated under degree-matched 5-fold cross-validation. Sampling and split designs vary by task as specified above.

Experiments and Results

Wallet-model and feature comparisons (RQ1)

Tree ensembles are competitive with the evaluated GNNs across the tested wallet benchmarks. Graph models show no consistent performance advantage, while structural-feature usefulness varies across networks.

Model comparison. Across all three networks, the models cluster tightly (Table 1; Figure 2). On the degree-matched Shiba benchmark (150 illicit and 1,332 negatives, 16 features), Random Forest reaches F1 0.799 ± 0.031 , Gradient Boosted Trees reach 0.812, GraphSAGE reaches 0.809, Plain GAT reaches 0.809, and Triple-Attention GAT reaches 0.823. The Triple-Attention GAT’s $+0.024$ edge falls completely inside the Random Forest cross-validation fold standard deviation (± 0.031). The fold standard deviation is descriptive and is not a paired equivalence test.

On the mature Bitcoin network, the ordering reverses: Random Forest wins outright at 0.935, GraphSAGE reaches 0.931, Plain GAT reaches 0.911, and the single-seed Triple-Attention GAT configuration scores lowest at 0.890; a separately documented multi-seed Triple-Attention result is approximately 0.912. On Dogecoin, which uses curated rather than degree-matched negatives, Random Forest reaches 0.993, Gradient Boost reaches 1.000, and the best reported GNN reaches 0.966.

Table 1: Wallet detector head-to-head under degree-matched 5-fold CV F1 (GATv2, 200 epochs, focal loss). SHIB and Bitcoin use degree-matched negative pools; DOGE uses curated negatives and therefore reflects an easier separation. The Bitcoin Triple-Attention entry is the single-seed result used in this head-to-head table.

Model	Type	Shiba F1	Bitcoin F1	DOGE F1
Random Forest	tabular, no graph	0.799 ± 0.031	0.935	0.993
Gradient-boosted tree	tabular, no graph	0.812	not reported	1.000

Model	Type	Shiba F1	Bitcoin F1	DOGE F1
GraphSAGE	graph	0.809	0.931	0.966
Plain GAT	graph	0.809	0.911	0.957
Triple-Attention GAT	graph	0.823	0.890	0.960
Logistic Regression	linear	0.478	not reported	not reported
best GNN – Random Forest		+0.024	−0.004	−0.027

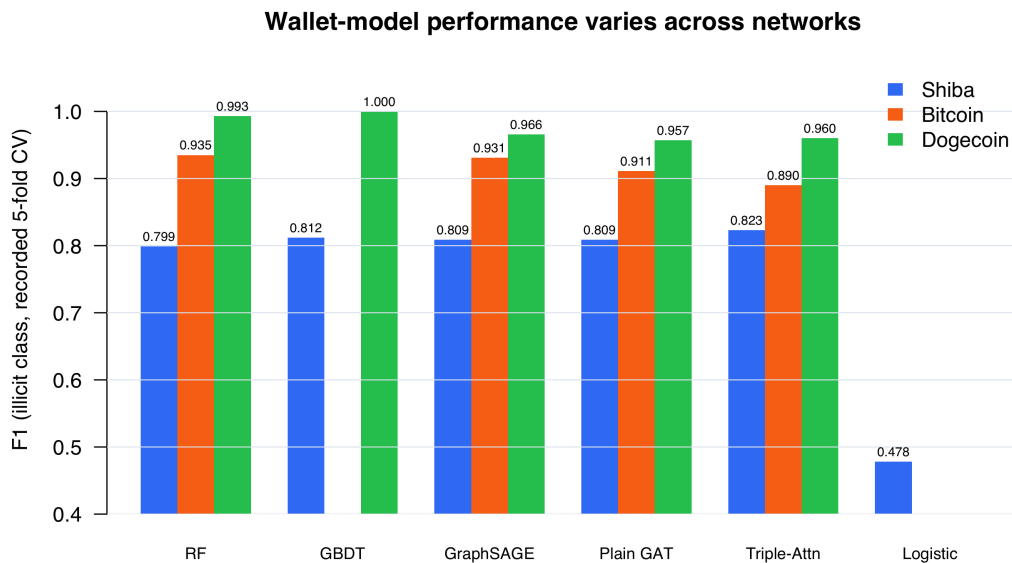


Figure 2: F1 scores for the recorded model configurations across three wallet benchmarks. Model ordering varies by network, and the evaluated GNNs show no consistent advantage over simple tree ensembles.

Feature importance and ablations. Impurity-based (Gini) feature importance reveals that the primary predictors on Shiba are purely behavioral: wallet age (0.294) and recency (0.173), whereas log PageRank accounts for only 0.059. Structural motifs (0.77 F1 alone) fail to beat static trees.

However, permutation-based importance places structural features such as k-core and PageRank higher, and matched ablations confirm that structural features carry network-specific value. On Bitcoin, Random Forest reaches F1 of approximately 0.893 using structural features alone, compared to 0.844 using behavioral features alone. On a distinct Bitcoin temporal task, the library-based EvolveGCN-O implementation reaches F1 of approximately 0.090; this result is not directly comparable with Table 1.

Why the Structural Signal Is Network-Specific

Structural-feature usefulness differs across networks, reflecting differences in degree concentration and network topology (Table 2). On the flatter Bitcoin network, where the top 1% of wallets hold 32% of connections, structural-only Random Forest reaches 0.893 (GNN 0.869). On Shiba (top 1% holds 48%), high-velocity exchange hubs mask illicit proximity; removing the five largest Shiba hubs roughly doubles illicit-versus-normal distance separation from 0.09 to 0.20 hops. The effect is small and non-monotone across pruning depths. On Dogecoin, where the top 1% of wallets hold 63% of connections, structural-only models achieve 0.754 (tree) and 0.629 (GNN).

Table 2: Network concentration and structural-only wallet-detection results across degree-matched 5-fold cross-validation.

Coin	Top-1% degree share	Struct-only Tree F1	Struct-only GNN F1
Bitcoin	32%	0.893	0.869
Shiba	48%	0.557	0.497
Dogecoin	63%	0.754	0.629

Forensic Traps Caught

1. **The Look-Ahead Leak:** Smearing connection lifetime backwards across dates produced an artificial 7% volatility forecasting gain in Dogecoin (0.934 MSFE ratio). Rebuilt causally, it collapsed to 1.044.
2. **The Label Leak:** Allowing test-set labels to influence distance-to-illicit metrics artificially inflated structural model scores.
3. **Price-Feed Sensitivity:** A Dogecoin predictive result appeared on one exchange feed and vanished on two others, demonstrating that it was a feed-level quoting artifact.

The Price Null (RQ2)

Across all primary feeds and out-of-sample tests, adding illicit-flow signals fails to improve market forecasts (Table 3). For Shiba, the HAR-RV-X MSFE ratio is 1.0025 ($p = 0.358$ in the reported Toda–Yamamoto / Granger causality test). For Ethereum, the MSFE ratio is 1.0944, while one in-sample specification retains a significant association (+0.135 SD, $p = 0.025$) without corresponding out-of-sample improvement. Dogecoin results are sensitive to the price feed.

Table 3: Price null, out-of-sample HAR-RV-X. SHIB and ETH are on the primary yfinance / Garman-Klass feed; DOGE is shown on its robustness feed (1.044), because on the primary feed DOGE is the documented artifact.

Coin	OOS RV MSFE	TY illicit→RV	→ret	Verdict
SHIB	1.0025	0.358	0.516	no improvement detected; limited OOS power
ETH	1.0944	0.111	0.255	OOS null; sig. in-sample coef. (+0.135 SD, $p = 0.025$)
DOGE	1.044	0.270	0.0007	documented price-feed artifact

When extending testing to market microstructure—Amihud illiquidity, Corwin-Schultz spreads, and hourly exchange-arrival pressure—apparent statistical associations fail to demonstrate temporal stability under rolling calibrations. For the Ethereum liquidity relationship, comparing the rolling-window parameter profile against calibration benchmarks reveals that the fraction of positive estimates is approximately 0.68 for the observed series, 0.49 for the shuffled null, and 0.93 for the injected signal (Figure 3). The observed profile does not exhibit the persistent directional sign expected of a genuine economic signal, clustering closer to the uninformative baseline.

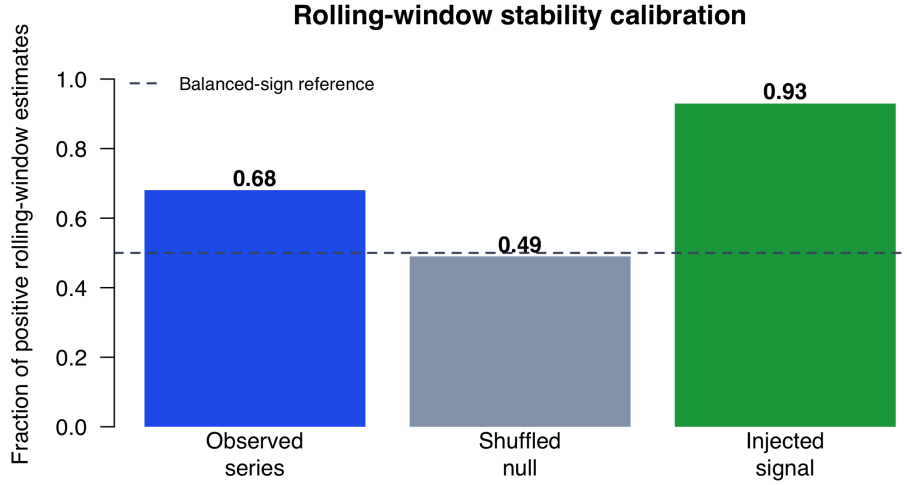


Figure 3: Fraction of positive rolling-window estimates for the observed Ethereum liquidity relationship and two calibration series. The observed profile does not establish a stable forecasting signal across the evaluated periods, failing the rolling calibration stability bar.

Additional Findings and Their Limits

Two supporting analyses produce positive but qualified associations:

1. **Miner Reserve Liquidations Pre-Merge:** Prior to the Ethereum Proof-of-Stake Merge (September 2022), weekly miner-reserve changes were associated with subsequent weekly ETH and BTC returns ($p < 0.01$): retention preceded stronger weeks, while structural distribution from mining pool treasury addresses preceded weaker weeks. The finding is modest and specification-sensitive; the calendar-grid result is weaker, the few-clusters test is borderline, and the formal break test is not significant. While operational decommissioning costs are a plausible economic mechanism, the analysis does not directly test decommissioning costs.
2. **Trade-Flow Imbalance and DEX Candidates:** High-frequency order flow imbalance on Binance BTCUSDT spot-trade data (January 15–17, 2024) shows a short-horizon predictive association on the held-out day across evaluated horizons of 1, 5, 30, and 300 seconds. Predictive alpha decays rapidly within 180–300 seconds, and the small directional edge is measured before trading fees and execution latency. Separately, our behavioral DEX screening procedure isolates net-flat, pool-dominating trader activity consistent with wash trading, while identifying remaining overlap with legitimate market-making and arbitrage.

The “Wash-Discount” (M-Score Analog)

Borrowing from the Beneish M-score used in forensic accounting to detect earnings manipulation, we formulate a candidate-associated “**Wash-Discount**” (WD_i) **metric** for decentralized exchange pools:

$$WD_i = 1 - \frac{\text{Volume}_{\text{organic}}}{\text{Volume}_{\text{reported}}} = \frac{\sum_{a \in \mathcal{C}_i} V_{a,i}}{V_{i,\text{reported}}}, \quad (2)$$

where $\mathcal{C}_i$ (or $\mathcal{E}_{\text{wash}}$) is the explicitly selected candidate-address set for pool i , and numerator and denominator use the same monetary-volume unit and observation window. The metric allows regulators and market aggregators (e.g., CoinMarketCap, CoinGecko) to restate reported pool volumes to true economic liquidity. However, flagged candidate volume is not automatically proven wash volume; classification requires token, liquidity, and trading-context validation.

Cross-Asset Relational Analysis

In a separate cross-asset analysis evaluating spillover graphs between Bitcoin, Ether, the S&P 500, VIX, and gold, contemporaneous correlations increase sharply during broad liquidity shocks. However, lagged cross-asset graph features provide zero incremental return predictability out of sample, reinforcing the Informational Efficiency hypothesis (Fama 1970; Grossman and Stiglitz 1980).

Detection as Triage, Not as an Alarm (RQ3)

The full-anchor EVM bytecode detector achieves an out-of-sample AUC of 0.876 (0.878 ± 0.002), with bootstrap 95% CI [0.840, 0.908]. Table 4 reports a ranking evaluation (AUC 0.8804) at 5.95% test prevalence, showing that the model concentrates true attackers $13.3\times$ above the base rate in the top 1% (Figure 4). These test-set precision figures should not be transferred directly to the rare target deployment population.

Table 4: Ranking value, rank-yes quantified (attacker detector, test base rate 5.95%).

Top- k	Precision	Attackers captured	Lift over base
Top 1%	0.79	13%	$13.3\times$
Top 5%	0.57	48%	$9.6\times$

Top- k	Precision	Attackers captured	Lift over base
Top 10%	0.36	61%	6.1×

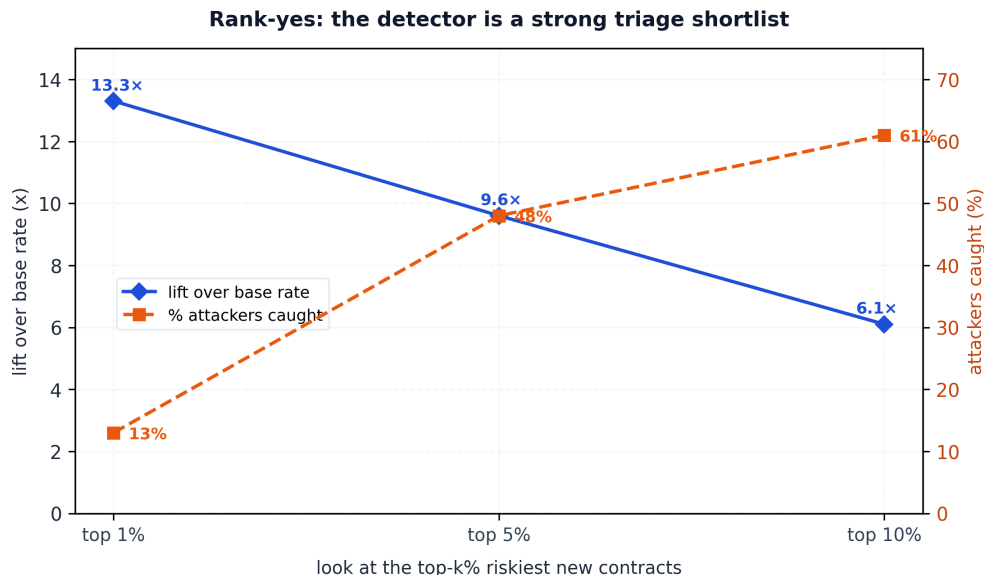


Figure 4: Ranking evaluation at 5.95% test prevalence. Reviewing the top 1%, 5%, and 10% of scored contracts captures approximately 13%, 48%, and 61% of attackers, respectively, confirming that the detector functions effectively as a triage shortlist.

Across the estimated attacker-prevalence band ($p \approx 2.8 \times 10^{-6}$ to 8.1×10^{-6} , central estimate 4.4×10^{-6}), the saved analysis projects approximately **40,700 to 116,700 benign reviews per catch** at the operating point targeting 90% recall. To maintain an affordable workload of fewer than ten benign reviews per catch at 0.1% prevalence and 90% recall would require an FPR below 0.00901, corresponding to an AUC exceeding 0.995 under an equal-variance binormal ROC model. Consequently, deploying the detector as an autonomous standalone alarm would generate an operational review workload far exceeding institutional capacity. Operational feasibility is strictly conditional on target prevalence, chosen classification thresholds, and available review resources.

Pre-filtering by mixer funding lifts AUC by only +0.008 and imposes a severe purity-recall trade-off, capturing only 33% of attackers while missing 67%.

The Deployment Gap Is General

Rare-event prevalence fundamentally determines operational review workload while operating-point error rates are held fixed (Tables 5–6; Figure 5). The deployment gap is not unique to smart-contract exploit detection; it is a universal mathematical property of rare-event classification across information systems.

Table 5: Projected review workload across blockchain-forensic tasks (reviews per catch at 90% recall where stated).

Task and configuration	F1	AUC	Projected Workload and Operational Context
Attacker contracts, full-anchor XGBoost	—	0.876	40,700–116,700 benign reviews/catch across estimated prevalence band ($p \approx 4 \times 10^{-6}$); 90% recall
Elliptic AML, recorded configuration	0.826	0.913	131–319 benign reviews/catch at 0.14–0.34% prevalence
Phishing, recorded configuration	0.911	0.989	15 benign reviews/catch in recorded scenario
Ponzi, Random Forest	0.738	0.959	117 benign reviews/catch at 0.1% prevalence; 90% recall
Rug pull, early-feature Random Forest	0.913	0.956	0.076 benign reviews/catch at recorded native prevalence

Table 6: Cross-domain review workload examples across varying base-rate environments. Operational feasibility depends on the specified review budget and target prevalence.

Dataset	Split/scenario	F1 or accuracy	Projected benign reviews/catch	Base-rate context
NSL-KDD intrusion	official Train+/Test+, canonical XGBoost	0.79	approximately 104–114	Moderate-prevalence benchmark
Spambase	random split, projected at 0.1% prevalence	0.947	approximately 20	Low-prevalence projection (0.1%)
ClaMP PE malware	random split, native prevalence	0.987	1.5	High-prevalence benchmark (balanced)

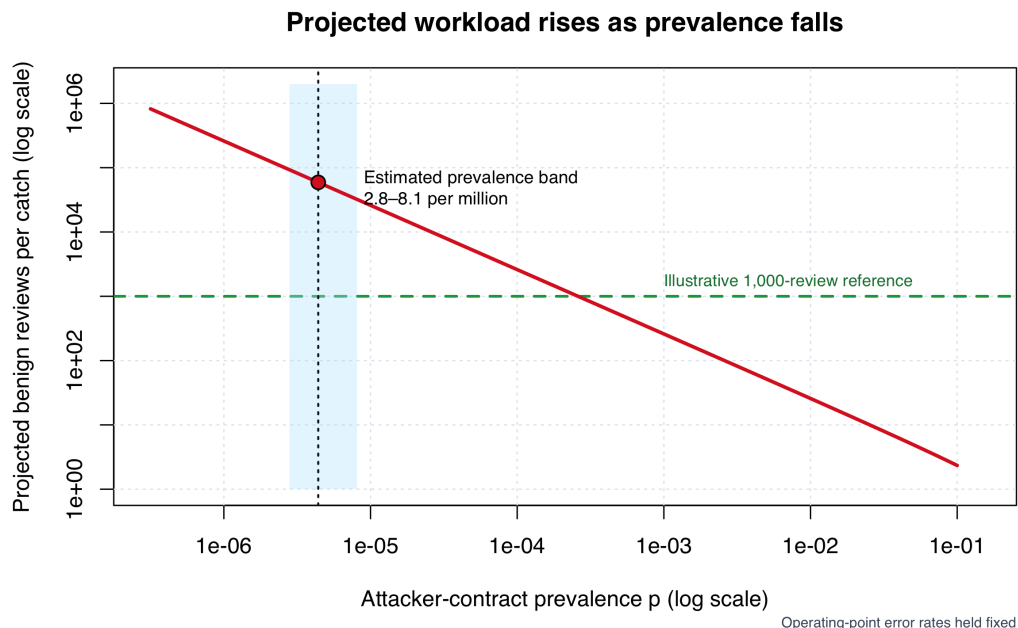


Figure 5: Projected benign reviews per catch as prevalence varies while evaluated operating-point error rates are held fixed. The shaded region marks the estimated attacker-contract prevalence band ($p \approx 4 \times 10^{-6}$); operational suitability depends on review budget and workflow.

Discussion & Stakeholder Impact

The blockchain intelligence industry is currently caught in a multi-billion dollar **“complexity trap.”** Despite heavy venture-capital investment into GNNs and “blackbox” AI architectures, our findings demonstrate that the marginal utility of these models over simple behavioral heuristics is effectively zero across multiple benchmarks. For institutional investors and quantitative trading desks, the “alpha” promised by on-chain illicit-flow signals is an illusion born of look-ahead leakage and feed artifacts. Financial institutions pay a substantial “complexity premium” for software that provides no better signal than basic transaction recency and volume metrics (Makarov and Schoar 2020).

Our empirical evaluations directly examine public labels and open-source implementations. While commercial vendors may incorporate proprietary intelligence feeds, private off-chain attribution databases, and specialized human investigator workflows, public case accounts (such as Bitfinex and Silk Road) illustrate that primary interdictions continue to rely heavily on behavioral errors, subpoenaed records, and conventional off-chain evidence. Commercial systems and their private off-chain datasets are not directly benchmarked here; rather, the burden of proof rests on vendors

to demonstrate that complex graph architectures provide incremental operational performance over simple baselines when evaluated using strictly point-in-time features.

The Staleness Crisis and Public-Label Cohort Activity

Public illicit labels exhibit rapid operational obsolescence and substantial dormancy within the evaluated cohorts (Table 7). In our examined Ethereum cohort, public labels exhibit an approximate **two-year activity half-life**, with only 12.9% of flagged addresses remaining active into 2024. In Dogecoin, the scam neighborhood became entirely dormant by 2022 ($\sim 0\%$ active into 2024).

A separately defined 2025+ comparison shows that OFAC-sanctioned addresses maintain higher longevity than scam addresses (26.5% vs. 13.9%, $z = 3.17$). This dormancy reflects the ephemeral operational lifecycle of criminal deposit addresses, single-use scam vectors, and intermediate wallets, which are routinely abandoned after illicit campaigns. Importantly, dormancy limits the ongoing coverage of static blacklists for real-time monitoring, though it does not invalidate historical labels. This rapid decay underscores the necessity of dynamic, entity-level re-identification rather than relying on static blacklist tags (Chainalysis 2024; TRM Labs 2024).

Table 7: Label staleness measured as per-address dormancy.

Dataset	Activity half-life	Active into 2024	Separate 2025+ comparison
ETH flagged ($n = 3,279$)	approximately 2 years	12.9%	OFAC 26.5% vs. ScamSniffer 13.9% ($z = 3.17$)
DOGE scam neighborhood	cohort largely dormant by 2022	approximately 0%	not evaluated

Stakeholder Impact & Practice: Who Benefits?

Our findings provide clear operational guidelines across three primary stakeholder groups:

1. **Regulators and Policymakers:** Mandates for automated transaction blocking under FATF Travel Rule frameworks (Financial Action Task Force 2021; Financial Crimes Enforcement

Network 2020b, 2020a) risk creating “security theater” if implemented without regard to base rates. Enforcing automated blocking at an attacker base rate of $p \approx 4 \times 10^{-6}$ imposes an intolerable operational tax of tens of thousands of false-positive reviews per legitimate catch. Regulators should mandate that screening evaluations report target-population prevalence, recall, false-positive rates, and expected review workloads. Automated models should serve strictly as **risk-triage shortlists** to prioritize manual human review rather than autonomous kill-switches.

2. **Cryptocurrency Exchanges and DEX Protocols:** Rather than relying on static blacklist tags, compliance teams should implement behavioral gating funnels and the **Wash-Discount (WD_i) metric** to restate reported liquidity and de-anonymize synthetic volume manipulation. Flagged candidate activity must be evaluated alongside token legitimacy, liquidity, market-making, and arbitrage context before volume is classified.
3. **Blockchain Analytics Developers:** Commercial vendors must address the **two-year half-life** of public labels. Pricing models and performance claims should reflect entity-level clustering, transaction recency, and real-time behavioral features rather than marketing unverified graph neural network complexity. Models must report provenance, observation windows, temporal coverage, simple baselines, and operating-point workloads conditional on target prevalence.

Conclusion and Future Work

This research establishes the empirical and operational limits of public illicit labels in blockchain forensics across four core dimensions:

1. **Network-Specific Structural Value (RQ1):** Simple tree ensembles on behavioral features ($F1 \approx 0.80$) match or outperform complex graph neural networks on degree-matched benchmarks. Structural features do not add consistent marginal value in top-heavy networks, but on flatter topologies like Bitcoin, structural-only Random Forest performance ($F1 \approx 0.893$) exceeds behavioral-only performance (0.844). Blanket claims of behavioral dominance or structural necessity fail to account for network concentration.
2. **Market Prediction Null (RQ2):** Illicit-flow measures provide no consistent forecasting improvement for returns, realized volatility, illiquidity, spreads, or exchange arrival. Apparent

signals are exposed as artifacts of look-ahead leaks, price-feed quoting discrepancies, or transient window variations failing rolling calibration stability tests.

3. **Operational Review Burden and the Base-Rate Fallacy (RQ3):** At an empirical attacker base rate of 2.8–8.1 per million deployments, rare-event mechanics impose a projected workload of approximately 40,700 to 116,700 benign reviews per catch at 90% recall. This high projected review burden—strictly conditional on target prevalence, classification thresholds, and institutional review capacity—precludes standalone automated alarms under realistic operating budgets, while supporting the detector’s value as a triage ranking tool ($13.3\times$ lift in the top 1%) to prioritize analyst review.
4. **DEX Wash-Trade Screening and Wash-Discount (RQ4):** The behavioral screening pipeline and the Wash-Discount score (WD_i) provide an actionable forensic mechanism for isolating pool-dominating, net-flat trading activity and restating reported volume, establishing consistent screening criteria while distinguishing suspicious volume from genuine market-making and arbitrage through contextual validation.

The path forward lies not in pursuing ever-heavier blackbox AI models, but in rigorous entity clustering that links visible entry points to the intermediary networks actually moving funds. Acknowledging the mathematical realities of the base-rate fallacy is essential to moving blockchain governance from expensive security theater toward effective, evidence-based forensic practice.

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